# SIMATS ENGINEERING

## ASSESSMENT TOOL 2 - Industry Problem Solving Task

**COURSE NAME: R PROGRAMMING** **COURSE CODE: ITA04**

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**COURSE OUTCOME COVERED**

**CO2:** R Programming Structures, Different types of loops such as: for (), while() loops, Control Statements, Looping Over Non-Vector Sets, - If-Else, Arithmetic and Boolean Operators and values, Default Values for Argument, Return Values, Deciding Whether to explicitly call return- Returning Complex Objects, Functions are Objective, No Pointers in R, Recursion. (BL4-BL5)

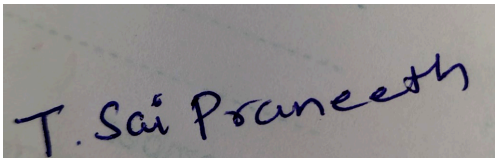
Assessment Tool Weightage: 25%

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***Code of Conduct:***

*I* T. Sai Praneeth (192525434) *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*




***Signature of the student:***

**INDUSTRY PROBLEM SOLVING TASK - ANSWERS**

Scenario: A retail company wants to use customer transaction data to understand spending, evaluate customer satisfaction, analyze associations, and improve marketing and loyalty decisions. The variables specified in the question paper include Customer_ID, Gender, Age, City, Product_Category, Purchase_Amount, Monthly_Income, Customer_Rating, Membership_Status, and Store_Type.

Important data note: The question paper specifies the fields but does not include the transaction dataset. Therefore, no numerical findings or test p-values are invented in this answer. The R code below is designed to calculate the results from the company's actual CSV file. Use the calculated values in the interpretation points when preparing the final submission.

**Initial Data Preparation in R**

Save the transaction file as retail_transactions.csv in the current R working directory. Run this preparation block once before running the code for Questions 1-10. The code checks the column names, converts numeric fields, and standardizes surrounding whitespace.

```
# Load the retail transaction dataset
retail <- read.csv("retail_transactions.csv",
                   stringsAsFactors = FALSE,
                   na.strings = c("", "NA", "N/A"))

# Check that every field required by the assignment is present
required_cols <- c("Customer_ID", "Gender", "Age", "City",
                   "Product_Category", "Purchase_Amount",
                   "Monthly_Income", "Customer_Rating",
                   "Membership_Status", "Store_Type")
missing_cols <- setdiff(required_cols, names(retail))
if (length(missing_cols) > 0) {
  stop(paste("Missing columns:", paste(missing_cols, collapse = ", ")))
}

# Convert numeric fields safely (also removes commas and currency symbols)
num_cols <- c("Age", "Purchase_Amount", "Monthly_Income", "Customer_Rating")
for (v in num_cols) {
  retail[[v]] <- as.numeric(gsub("[$, ]", "", as.character(retail[[v]])))
}

# Remove leading/trailing spaces from text fields
text_cols <- c("Gender", "City", "Product_Category",
               "Membership_Status", "Store_Type")
for (v in text_cols) {
  retail[[v]] <- trimws(as.character(retail[[v]]))
}

# Inspect the structure and missing values
str(retail)
summary(retail)
colSums(is.na(retail))
```

Data preparation explanation: Missing values are not automatically replaced with zero because zero may be a genuine value and would distort the statistics. Each analysis uses the available valid observations for the variables being studied. Confirm that Purchase_Amount, Monthly_Income, and Customer_Rating are numeric after import.

**Question 1: Measures of Central Tendency for Purchase Amount**

Answer: The mean, median, and mode describe different aspects of customer spending. The mean is the sum of all valid purchase amounts divided by the number of purchases. The median is the middle value after sorting the amounts. The mode is the most frequently observed purchase amount. Together, these measures help management understand typical spending and whether unusually large purchases influence the average.

```
purchase <- retail$Purchase_Amount
purchase <- purchase[is.finite(purchase)]

# A statistical mode function; reports if all values occur only once
get_mode <- function(x) {
  x <- x[!is.na(x)]
  if (length(x) == 0) return(NA_character_)
  frequencies <- table(x)
  if (max(frequencies) == 1) return("No repeated value")
  names(frequencies)[frequencies == max(frequencies)]
}

cat("Number of valid purchases:", length(purchase), "\n")
cat("Mean purchase amount:", mean(purchase), "\n")
cat("Median purchase amount:", median(purchase), "\n")
cat("Mode(s) of purchase amount:", paste(get_mode(purchase), collapse = ", "),
"\n")
cat("Skewness clue - mean minus median:", mean(purchase) - median(purchase), "\n")

hist(purchase, main = "Distribution of Purchase Amount",
     xlab = "Purchase Amount", col = "white", border = "black")
abline(v = mean(purchase), lty = 2, lwd = 2)
abline(v = median(purchase), lty = 1, lwd = 2)
```

Interpretation and recommendation: Compare the mean and median produced by the code. If the mean is substantially greater than the median, a small number of high-value purchases may be pulling the mean upward; if the mean is substantially smaller, unusually low values may be influencing it. For a skewed retail spending distribution or one containing outliers, the median is usually the best single measure of typical customer spending because it is resistant to extreme purchases. If the distribution is approximately symmetric and contains no important outliers, the mean is also a suitable summary. The mode is useful for identifying a frequently occurring price or transaction value, but it may be uninformative when purchase amounts are mostly unique.

**Question 2: Measures of Dispersion in Customer Spending**

Answer: Dispersion measures show how widely customer purchase amounts vary. The range is the difference between the maximum and minimum. Variance measures average squared deviations from the mean, standard deviation expresses spread in the same units as the purchase amount, and the interquartile range (IQR) measures the spread of the middle 50% of purchases. The coefficient of variation (CV) is included as a relative measure to compare dispersion against the average spending level.

```
purchase <- retail$Purchase_Amount
purchase <- purchase[is.finite(purchase)]

spending_range <- diff(range(purchase))
spending_variance <- var(purchase)       # sample variance
spending_sd <- sd(purchase)              # sample standard deviation
spending_iqr <- IQR(purchase)            # Q3 - Q1
spending_cv <- (spending_sd / mean(purchase)) * 100

cat("Minimum:", min(purchase), "\n")
cat("Maximum:", max(purchase), "\n")
cat("Range:", spending_range, "\n")
cat("Variance:", spending_variance, "\n")
cat("Standard deviation:", spending_sd, "\n")
cat("First quartile:", quantile(purchase, 0.25), "\n")
cat("Third quartile:", quantile(purchase, 0.75), "\n")
cat("IQR:", spending_iqr, "\n")
cat("Coefficient of variation (%):", spending_cv, "\n")

boxplot(purchase, horizontal = TRUE, main = "Spread of Purchase Amounts",
        xlab = "Purchase Amount", col = "white", border = "black")
```

Interpretation: A large range indicates that minimum and maximum spending differ widely, but it can be strongly affected by outliers. Standard deviation describes overall variability; IQR is more resistant to unusually large or small purchases. A relatively large CV indicates that spending varies considerably compared with the mean, while a lower CV suggests more consistent spending. There is no universal CV threshold for all retail datasets, so judge consistency in business context and compare groups or time periods if possible. If the boxplot shows many points far beyond the whiskers, investigate extreme values before making decisions.

**Question 3: Exploratory Analysis of Customer Rating by Membership Status**

Answer: Customer_Rating should be summarized using the number of observations, mean, median, standard deviation, minimum, and maximum for each Membership_Status. A boxplot is appropriate for comparing the rating distributions of Premium and Regular members because it displays the median, quartiles, spread, and potential outliers. The comparison should consider both typical rating and variation, not only the average.

```
rating_data <- retail[!is.na(retail$Customer_Rating) &
                      !is.na(retail$Membership_Status), ]
rating_groups <- split(rating_data$Customer_Rating,
                       rating_data$Membership_Status)

rating_summary <- do.call(rbind, lapply(rating_groups, function(x) {
  c(n = length(x), mean = mean(x), median = median(x),
    sd = sd(x), min = min(x), max = max(x))
}))
print(round(rating_summary, 3))

boxplot(Customer_Rating ~ Membership_Status, data = rating_data,
        main = "Customer Ratings by Membership Status",
        xlab = "Membership Status", ylab = "Customer Rating",
        col = "white", border = "black")
stripchart(Customer_Rating ~ Membership_Status, data = rating_data,
           vertical = TRUE, method = "jitter", add = TRUE,
           pch = 16, cex = 0.6)
```

Interpretation: Compare the Premium and Regular rows of the summary table. A higher median or mean rating for Premium members would be consistent with greater satisfaction among that group, while a narrower spread would indicate more consistent ratings. Similar medians and overlapping boxplots suggest that the descriptive difference may be small. This is exploratory analysis only: a visual difference does not by itself prove statistical significance or that membership caused the difference. Confirm the exact categories present in the dataset before interpreting the comparison.

**Question 4: Relationship Between Monthly Income and Purchase Amount**

Answer: Bivariate analysis examines two numeric variables together. A scatter plot can reveal a positive relationship (higher income tends to accompany higher purchase amounts), a negative relationship, or little apparent linear pattern. Pearson correlation measures linear association; Spearman correlation measures monotonic association based on ranks and can be more appropriate when the pattern is not linear or is affected by outliers.

```
income_spend <- retail[complete.cases(retail[, c("Monthly_Income",
                                                 "Purchase_Amount")]), ]

plot(income_spend$Monthly_Income, income_spend$Purchase_Amount,
     main = "Monthly Income vs Purchase Amount",
     xlab = "Monthly Income", ylab = "Purchase Amount",
     pch = 16, col = "black")
abline(lm(Purchase_Amount ~ Monthly_Income, data = income_spend),
       lty = 2, lwd = 2)

pearson_result <- cor.test(income_spend$Monthly_Income,
                           income_spend$Purchase_Amount,
                           method = "pearson")
spearman_result <- cor.test(income_spend$Monthly_Income,
                            income_spend$Purchase_Amount,
                            method = "spearman", exact = FALSE)
print(pearson_result)
print(spearman_result)
```

Interpretation: A positive correlation means higher monthly income tends to be associated with higher purchase amounts; a negative correlation means the variables tend to move in opposite directions. A correlation near zero indicates little linear association, not necessarily no relationship. As a rough guide, the absolute correlation may be described as weak, moderate, or strong, but the context and scatter plot matter more than rigid cutoffs. Management may use the result to tailor product bundles or promotions by spending capacity, but income should not be treated as the sole predictor, and correlation does not establish causation.

**Question 5: Independent Samples T-Test for Male and Female Customers**

Answer: An independent two-sample Welch t-test can test whether male and female customers have different mean purchase amounts. It does not require the two groups to have equal variances. Use the following hypotheses at a significance level of 0.05: H0: the male and female population mean purchase amounts are equal; H1: the means are different.

```
gender_data <- retail[!is.na(retail$Gender) &
                      !is.na(retail$Purchase_Amount), ]
gender_data$Gender <- tolower(trimws(gender_data$Gender))
gender_data <- subset(gender_data, Gender %in% c("male", "female"))
gender_data$Gender <- factor(gender_data$Gender,
                             levels = c("female", "male"))

print(aggregate(Purchase_Amount ~ Gender, data = gender_data,
                FUN = function(x) c(n = length(x), mean = mean(x),
                                    median = median(x), sd = sd(x))))

t_result <- t.test(Purchase_Amount ~ Gender,
                   data = gender_data, var.equal = FALSE,
                   alternative = "two.sided")
print(t_result)

if (t_result$p.value < 0.05) {
  cat("Decision: Reject H0 at the 5% significance level.\n")
  cat("There is evidence of a difference in mean purchase amount.\n")
} else {
  cat("Decision: Do not reject H0 at the 5% significance level.\n")
  cat("The sample does not provide sufficient evidence of a mean difference.\n")
}
```

Interpretation and business use: Report the mean purchase amount for each gender, the t statistic, degrees of freedom, p-value, and 95% confidence interval from the R output. If $p < 0.05$, the analysis provides evidence of a mean difference; inspect the group means to identify which group spends more on average. If $p >= 0.05$, there is insufficient evidence of a difference, which is not proof that the means are exactly equal. Marketing should not be designed from gender alone; combine the result with product preferences, income, customer needs, and fairness considerations.

**Question 6: ANOVA of Customer Ratings Across Store Types**

Answer: One-way analysis of variance (ANOVA) compares mean Customer_Rating across Urban, Semi-Urban, and Rural store types. The hypotheses are H0: all three population means are equal; H1: at least one mean differs. ANOVA is preferred to repeatedly running pairwise t-tests because it evaluates the groups together while controlling the overall test approach.

```
store_rating <- retail[!is.na(retail$Customer_Rating) &
                       !is.na(retail$Store_Type), ]
store_rating$Store_Type <- factor(store_rating$Store_Type)

print(aggregate(Customer_Rating ~ Store_Type, data = store_rating,
                FUN = function(x) c(n = length(x), mean = mean(x),
                                    median = median(x), sd = sd(x))))

anova_model <- aov(Customer_Rating ~ Store_Type, data = store_rating)
anova_table <- summary(anova_model)
print(anova_table)

p_anova <- anova_table[[1]][["Pr(>F)"]][1]
if (!is.na(p_anova) && p_anova < 0.05) {
  cat("At least one store-type mean rating differs significantly.\n")
  print(TukeyHSD(anova_model))
} else {
  cat("No statistically significant mean-rating difference was detected.\n")
}

# Diagnostic plots for checking model assumptions
par(mfrow = c(1, 2))
plot(anova_model, which = 1)
plot(anova_model, which = 2)
par(mfrow = c(1, 1))
```

Interpretation: If the ANOVA p-value is below 0.05, reject H0 and use the Tukey HSD output to determine which pairs of store types differ. If the p-value is at least 0.05, there is insufficient evidence that mean ratings differ. Check residual plots for major non-normality patterns or unequal variance. If assumptions are badly violated, consider a suitable robust or non-parametric alternative. The business response should focus on the store types and pairs with confirmed differences, for example by investigating service quality, product availability, wait times, or staffing in lower-rated locations.

**Question 7: Chi-Square Test of Membership Status and Product Category**

Answer: A chi-square test of independence determines whether Membership_Status and Product_Category are associated. H0: membership status and product category preference are independent; H1: they are associated. The contingency table gives counts for every combination. Row percentages help compare each membership group's category mix.

```
association_data <- retail[!is.na(retail$Membership_Status) &
                           !is.na(retail$Product_Category), ]
association_table <- table(association_data$Membership_Status,
                           association_data$Product_Category)
print(association_table)
print(round(prop.table(association_table, margin = 1) * 100, 1))

chi_initial <- chisq.test(association_table, correct = FALSE)
if (all(dim(association_table) == c(2, 2)) &&
    any(chi_initial$expected < 5)) {
  association_test <- fisher.test(association_table)
} else if (any(chi_initial$expected < 5)) {
  association_test <- chisq.test(association_table,
                                 simulate.p.value = TRUE, B = 10000)
} else {
  association_test <- chi_initial
}
print(association_test)
print(round(chi_initial$stdres, 2))  # standardized residuals for category patterns

# Cramer's V measures association strength (0 to 1)
n_total <- sum(association_table)
cramers_v <- sqrt(as.numeric(chi_initial$statistic) /
                  (n_total * min(nrow(association_table) - 1,
                                 ncol(association_table) - 1)))
cat("Cramer's V:", cramers_v, "\n")
barplot(prop.table(association_table, margin = 1), beside = FALSE,
        legend.text = TRUE, xlab = "Membership Status",
        ylab = "Proportion", main = "Product Category Mix by Membership")
```

Interpretation: If the chosen test's p-value is below 0.05, reject H0 and conclude that the sample provides evidence of an association between membership and product category. Review row percentages and standardized residuals from the ordinary chi-square output to identify which combinations contribute most to the pattern. Cramer's V helps describe the strength of the association; statistical significance alone does not guarantee a large or commercially important effect. If p >= 0.05, there is insufficient evidence of an association. Use the category mix to personalize retention offers only where a meaningful, reliable pattern is present.

**Question 8: Identify a Priority Segment for a New Loyalty Program**

Answer: Exploratory data analysis should identify a segment that combines commercial value, customer count, and potential to improve retention. A practical first candidate is regular or non-premium customers who already show relatively high purchase amounts and satisfactory ratings in product categories with a sizeable customer base. These customers may have meaningful potential to upgrade or engage more frequently. This is a candidate strategy rather than a measured finding; the actual highest-priority segment must be selected from the calculated summaries below.

```
segment_data <- retail[!is.na(retail$Membership_Status) &
                       !is.na(retail$Product_Category), ]
segment_key <- interaction(segment_data$Membership_Status,
                           segment_data$Product_Category, drop = TRUE)
segment_groups <- split(segment_data, segment_key)

segment_summary <- do.call(rbind, lapply(segment_groups, function(g) {
  data.frame(
    Membership_Status = as.character(g$Membership_Status[1]),
    Product_Category = as.character(g$Product_Category[1]),
    transactions = nrow(g),
    unique_customers = length(unique(g$Customer_ID[!is.na(g$Customer_ID)])),
    mean_purchase = mean(g$Purchase_Amount, na.rm = TRUE),
    median_purchase = median(g$Purchase_Amount, na.rm = TRUE),
    mean_rating = mean(g$Customer_Rating, na.rm = TRUE)
  )
}))

# Candidate upgrade / engagement groups: exclude Premium members
candidates <- subset(segment_summary,
                     !grepl("premium", Membership_Status, ignore.case = TRUE))
candidates <- candidates[order(-candidates$mean_purchase,
                               -candidates$mean_rating,
                               -candidates$unique_customers), ]
print(head(candidates, 10))
```

Selection method: Review the top rows alongside transaction counts and unique-customer counts. Prefer a segment that is not just high-spending because of one or two transactions; check median spending and group size as well. A high mean rating may indicate that customers already have a positive experience, making a loyalty offer a plausible way to increase repeat visits. Management should pilot the offer with a measurable comparison group and track redemption, repeat-purchase rate, average order value, and cost per retained customer. Do not claim that a specific segment ranks first until the code has been run on the actual dataset.

**Question 9: Key Factors Associated with High Purchase Amounts**

Answer: Define high purchase amounts using a transparent rule, such as transactions above the 75th percentile. Then compare spending by Product_Category, Membership_Status, Store_Type, Gender, and City, and examine the bivariate relationship between Monthly_Income and Purchase_Amount. A multiple linear regression can be used as an additional exploratory check to examine whether selected variables remain associated with spending when considered together.

```
valid_purchase <- retail[!is.na(retail$Purchase_Amount), ]
q75 <- as.numeric(quantile(valid_purchase$Purchase_Amount, 0.75,
                           na.rm = TRUE))
valid_purchase$Purchase_Level <- ifelse(
  valid_purchase$Purchase_Amount > q75, "High", "Other")
cat("75th-percentile threshold for high purchase:", q75, "\n")
print(table(valid_purchase$Purchase_Level))

# Compare spending across important categorical variables
print(aggregate(Purchase_Amount ~ Product_Category, data = valid_purchase,
                FUN = function(x) c(n = length(x), mean = mean(x),
                                    median = median(x), sd = sd(x))))
print(aggregate(Purchase_Amount ~ Membership_Status, data = valid_purchase,
                FUN = function(x) c(n = length(x), mean = mean(x),
                                    median = median(x))))
print(aggregate(Purchase_Amount ~ Store_Type, data = valid_purchase,
                FUN = function(x) c(n = length(x), mean = mean(x),
                                    median = median(x))))

# Bivariate relationship with income
income_spend <- valid_purchase[!is.na(valid_purchase$Monthly_Income), ]
print(cor.test(income_spend$Monthly_Income,
               income_spend$Purchase_Amount, method = "spearman",
               exact = FALSE))

# Optional adjusted analysis of several factors together
model_data <- retail[, c("Purchase_Amount", "Monthly_Income", "Age", "Gender",
                         "City", "Product_Category", "Customer_Rating",
                         "Membership_Status", "Store_Type")]
model_data <- na.omit(model_data)
spending_model <- lm(Purchase_Amount ~ Monthly_Income + Age + Gender + City +
                     Product_Category + Customer_Rating + Membership_Status +
                     Store_Type, data = model_data)
summary(spending_model)
```

Interpretation: The group summaries show which categories, membership groups, and store types have higher average and median purchases. The income correlation indicates the direction and strength of the monotonic association. In the regression output, coefficients describe adjusted associations relative to reference categories or per-unit changes in numeric predictors; p-values and confidence intervals help assess uncertainty. Factors should be called “influential” only in the predictive or statistical sense, not as proven causes. Check sample sizes and unusually influential observations before using the results in business decisions.

**Question 10: Business Analytics Report and Actionable Recommendations**

Answer: The final report should combine descriptive measures with inferential test results and translate them into actions. Once the R code has been run on the transaction file, insert the actual statistics, test outputs, and group names in the report template below. The conclusions must follow the results rather than assuming that any comparison is significant.

**Report template**

Objective: To use exploratory data analysis and statistical testing to understand retail spending, customer satisfaction, income-spending patterns, and the relationship between membership and product preferences.

Spending profile: Report the number of valid purchases, mean, median, mode (if meaningful), standard deviation, and IQR of Purchase_Amount. Explain whether the spending distribution is skewed and choose the median or mean as the main measure of typical spending accordingly.

Customer satisfaction: Compare rating means, medians, and spread by Membership_Status and Store_Type. Report the ANOVA p-value and, if significant, the relevant Tukey pairwise comparisons. State plainly whether the data provides sufficient evidence of differences in mean ratings.

Customer group comparison: Report the male and female group means, Welch t-test statistic, degrees of freedom, p-value, and confidence interval. State whether the mean difference is statistically significant at alpha = 0.05 and describe its direction only after checking the group means.

Product preference and membership: Present the membership-by-category contingency table, row percentages, selected test, p-value, and Cramer's V when calculable. Describe which category patterns differ most, while distinguishing a statistically significant association from a large business effect.

Income and purchase behavior: Report the Pearson or Spearman correlation, its direction, magnitude, and p-value. Use the scatter plot to discuss whether the association is linear, monotonic, or influenced by outliers. State that association does not prove causation.

Loyalty-program priority: Identify the highest-potential non-premium segment from the segment summary, citing customer count, mean and median purchase amount, and rating. Avoid selecting a segment using mean purchase amount alone.

**Actionable recommendations**

- Use the median to describe typical spending when the purchase distribution is skewed or contains high-value outliers; retain the mean as a complementary measure for revenue planning.
- Develop offers around product categories and membership groups only when the EDA and association test show practically meaningful differences.

- If store-type ratings differ significantly, investigate operational causes in lower-rated store types and monitor ratings after service improvements.
- If income and spending are positively associated, test value-based bundles and appropriate price tiers; use multiple customer attributes rather than income alone.
- Pilot the loyalty program with a defined target segment and a comparison group. Evaluate repeat purchases, incremental spending, offer redemption, retention, and campaign cost before expanding it.
- Repeat the analysis periodically, validate missing values and outliers, and protect customer information when preparing reports or segment-level campaigns.

Overall conclusion: The appropriate business strategy should be based on the statistics calculated from the real transaction data. EDA describes the patterns, while the t-test, ANOVA, and chi-square test help assess specific hypotheses. Management should prioritize effects that are statistically supported, commercially meaningful, and validated through a measured pilot.

## RUBRICS FOR CALCULATION

| Industry Problem Solving Task | | | | | |
|---|---|---|---|---|---|
| **Criteria** | **Weightage** | **Level 4 – Exemplary** | **Level 3 – Proficient** | **Level 2 – Developing** | **Level 1 – Beginning** |
| Application of Techniques | 25 | Correctly applies central tendency, dispersion, and bivariate analysis techniques. | Applies most EDA techniques correctly with minor omissions. | Applies basic EDA techniques with limited accuracy. | Demonstrates limited or incorrect application of EDA techniques. |
| Statistical Analysis & Testing | 25 | Correctly performs and interprets T-test, ANOVA, and Chi-square analysis. | Performs statistical tests with minor interpretation errors. | Performs basic statistical analysis with limited interpretation. | Demonstrates poor understanding of statistical tests. |
| Problem Analysis & Data Interpretation | 20 | Accurately analyzes data patterns and derives meaningful insights. | Provides reasonable analysis with adequate interpretation. | Provides limited analysis and interpretation. | Analysis is weak or incorrect. |
| Decision-Making & Business Recommendations | 20 | Provides data-driven recommendations that effectively address the business problem. | Recommendations are relevant and supported by analysis. | Recommendations are basic with limited justification. | Recommendations are weak, unsupported, or missing. |
| Organization & Completeness | 10 | Response is well-structured, complete, and addresses all requirements. | Mostly complete and organized. | Partially complete with some missing components. | Incomplete and poorly organized. |